# Kwok Wai Taifa

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## Education

**Manipal University Jaipur** 2024 – 2028
*B.Tech, Computer Science Engineering (Artificial Intelligence & Machine Learning)* CGPA: 8.5 / 10
*Coursework:* Data Structures & Algorithms, Operating Systems, Database Management Systems, Statistics, Probabilistic AI.

## Experience

**AI Developer Intern** Jan 2026 – Present
*Banao Technologies — Software & AI Development* Jaipur, India

- **Sole author of NERVE**, the deterministic zero-LLM orchestration backbone and only always-on service of a **5-agent** platform: an async FastAPI/`asyncpg` event bus with a transactional outbox and reaper, per-agent circuit breakers, dead-letter digests and cron scheduling across **7 schema migrations** — cut cross-agent orchestration latency **35%** and proved **at-least-once delivery does not duplicate** rows under redelivery.
- Stopped provider-outage cascades by designing NERVE's failure taxonomy: a per-error-class exponential-backoff matrix (`agent_down` 3 retries at 30/120/600 s; `quota_exceeded` **0 retries, immediate abort**) and a quota monitor that marks an exhausted LLM provider down *globally* and skips its scheduled jobs, eliminating retry storms and wasted spend.
- **Sole author of META**, the intelligence layer converting raw cross-platform activity into per-person alignment via a **6-stage funnel** (raw → identity resolution → daily fact atoms → goals → deterministic flag engine → live star-plot API), with an **identity spine** unifying GitHub, Slack, email and metered-LLM telemetry onto one employee ID, and idempotent ingestion so per-person token/cost attribution survives retries.
- Built **Seller Radar** end to end across **3 production repos** (Django/React intranet, agent platform, Node meet-recorder), replacing a hand-filled sales scorecard with a pipeline that turns a recorded call into an evidence-linked, timestamped score — reconciling two independently-built scoring systems sharing no schema, identity model or rubric. **81 new tests**, **166 existing** kept green.
- Closed a **verified privilege-escalation vulnerability** letting a sales rep rewrite their own performance record (`PATCH` a 2/10 to a 9/10, `DELETE` a bad call, `POST` a fresh 10/10), by restricting writes to admins, retaining overrides with actor and timestamp, and soft-deactivating deletes so reporting denominators stay correct.
- Made the manager team heatmap **O(1) in headcount — 24 DB queries** → **2** at 12 reps, per-rep pattern analysis **11** → **6** flat in history size — and cut LLM hallucination ∼**40%** at <**120 ms p95** with a FAISS-backed RAG pipeline that **discards ungrounded scores** rather than emit a number a user cannot inspect.

## Projects

**Jarvis — Fully Offline On-Device Voice Assistant** [GitHub] 2026
*llama.cpp (CUDA), Qwen2.5-7B GGUF, faster-whisper, ONNX Runtime, ChromaDB, Python*

- Ran a **7B-parameter LLM at 34 tok/s on a 6 GB consumer GPU**, all **29/29 layers** offloaded, by serving a Q4_K_M GGUF in-process through llama.cpp — **zero cloud API calls** in the loop.
- Held **4–7 s end-to-end voice turns** by profiling the VRAM/context curve (chose `n_ctx`=8192; 32768 *silently* spilled KV cache to system RAM, −**13% throughput**) and pinning Whisper to CPU at **RTF 0.25** — partial GPU offload cost **34** → **10–15 tok/s** to save 1.3 s of one-shot STT.
- Shrank the embedding stack ∼**2.5 GB** → ∼**80 MB** (PyTorch → ONNX Runtime) at identical vectors; blocked **13/13** path-traversal attacks and **21/21** tool-safety checks; shipped behind a **200+ assertion** suite (**100%** tool-call accuracy, **27/27** grounding, **16/16** cross-restart memory).

**Verity — Multimodal Disinformation Intelligence Platform** [GitHub] 2026
*PyTorch, DistilBERT, SigLIP-2, NLI Cross-Encoders, ONNX Quantization, Neo4j*

- Fused **3 model families** — DistilBERT text forensics, SigLIP-2 AI-image/deepfake detection, NLI cross-encoder claim verification against Wikidata — into one credibility score with **sentence-level attribution**, making every verdict explainable rather than black-box.
- Cut model size ∼**4**× via **ONNX INT8 quantization** at held accuracy, closing a human-in-the-loop active-learning loop where analyst labels on low-confidence scans trigger transfer-learning retraining with early stopping and checkpoint recovery.

**AI-Growth — Agentic Commerce Guardrail System** [GitHub] 2026
*LangGraph, FastAPI, DynamoDB, Anthropic API, Python*

- Achieved **100% policy compliance** on transaction limits and **zero budget-overspend races** under concurrent load, by separating an untrusted LLM proposal layer from a deterministic gatekeeper with optimistic-locking concurrency control.
- Guaranteed tamper-evidence at <**120 ms** audit overhead via a **SHA-256 hash-chained** immutable ledger with automated budget-expiration enforcement.

**Hiring Agent — Rubric-Based Candidate Ranking** [GitHub] 2026
*Sentence-Transformers, FAISS, Pydantic, Python*

- Ranked **1,000+ applications in <8 s per batch** at **91% alignment** with human evaluators, by scoring FAISS embedding similarity against a rubric with schema-constrained Pydantic extraction, tuned on a hand-labeled validation set.

## Technical Skills

**Languages:** Python, C, SQL, JavaScript/TypeScript
**ML & Deep Learning:** PyTorch, scikit-learn, Transformers, Sentence-Transformers, DistilBERT, SigLIP-2, Cross-Encoders, Transfer Learning, Quantization (ONNX INT8), Whisper
**LLM Systems:** llama.cpp / GGUF, Local Inference & GPU Offload, RAG, LangGraph, FAISS, ChromaDB, Structured Extraction (Pydantic), Prompt-Injection Hardening
**Backend & Infra:** FastAPI, Django, React, Async Python, PostgreSQL, DynamoDB, Neo4j, Docker, AWS EC2, Cloudflare R2, CUDA, Event-Driven Architecture, Git
**Fundamentals:** Data Structures, Algorithms, Operating Systems, Statistics

## Leadership & Achievements

**Founder & President, Martial Arts & Self Defence Club — Manipal University Jaipur** 2025 – 2026

- Founded the university's first martial arts club, scaling it to **200+ active students** across 2 semesters. **Gold Medalist**, District Boxing Championship; **Bronze Medalist**, State Level; National-Level Participant via district and state qualification.